



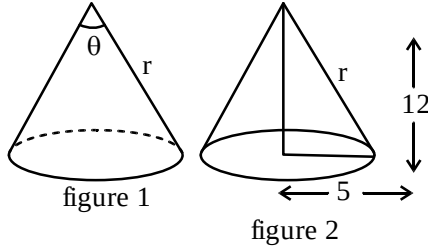
Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-06

- The maximum value of the function $f(x) = \frac{x}{8} + \frac{2}{x}$ on interval $[1, 6]$ is
 - 1
 - $\frac{9}{8}$
 - $\frac{13}{12}$
 - $\frac{17}{8}$
- A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. The lengths of the sides of the rectangular sheet are
 - 24
 - 32
 - 45
 - 60
- A wire of length 2 units is cut into two parts which are bent respectively to form a square of side x units and a circle of radius r units. If the sum of the areas of the square and the circle so formed is minimum, then
 - $2x(\pi + 4)r$
 - $(4 - \pi)x = \pi r$
 - $x = 2r$
 - $2x = r$
- The minimum distance of a point on the curve $y = x^2 - 4$ from origin, is
 - $\frac{\sqrt{5}}{2}$
 - $\frac{\sqrt{17}}{2}$
 - $\frac{\sqrt{15}}{2}$
 - $\frac{\sqrt{19}}{2}$
- A cone of maximum volume is inscribed in a given sphere, then ratio of the height of the cone to diameter of the sphere is
 - $\frac{2}{3}$
 - $\frac{3}{4}$
 - $\frac{1}{3}$
 - $\frac{1}{4}$
- A wire of length 28 cm is bent to form a circular sector, then the radius (in cm) of the circular sector such that the area of the circular sector is maximum is equal to
 - 5
 - 6
 - 7
 - 8
- A rectangle is inscribed in an equilateral triangle of side length $2a$ units. The maximum area of this rectangle can be:
 - $\sqrt{3}a^2$
 - $\frac{\sqrt{3}a^2}{4}$
 - a^2
 - $\frac{\sqrt{3}a^2}{2}$
- If atmosphere pressure at a height of h units is given by the function $P(h) = he^{-h}$ then pressure is maximum at the height of:
 - 1 unit
 - $\frac{1}{\sqrt{e}}$ units
 - e units
 - 2 units
- The maximum value of $\left(\frac{1}{x}\right)^{2x^2}$ is:
 - e
 - $e^{\left(\frac{1}{e}\right)}$
 - 1
 - e^e

10. A piece of paper in the shape of a sector of a circle (see figure 1) is rolled up to form a right – circular cone (see figure 2). The value of angle θ is:



- (1) $\frac{10\pi}{13}$ (2) $\frac{9\pi}{13}$
 (3) $\frac{5\pi}{13}$ (4) $\frac{6\pi}{13}$
11. The largest area of a trapezium inscribed in a semi-circle of radius R , if the lower base is on the diameter, is:

- (1) $\frac{3\sqrt{3}}{4}R^2$ (2) $\frac{\sqrt{3}}{2}R^2$
 (3) $\frac{3\sqrt{3}}{8}R^2$ (4) R^2

12. A window is in the form of a rectangle surmounted by a semi-circle. The total area of window is fixed. What should be the ratio of the areas of the semi-circular part and the rectangular part so that the total perimeter is minimum?

- (1) $\frac{\pi}{4}$
 (2) $\frac{1}{4}$
 (3) $\frac{\pi}{2}$
 (4) $\frac{1}{2}$

13. Determine the absolute extrema for the following function and interval. $g(t) = 2t^3 + 3t^2 - 12t + 4$ on the interval $t \in [0, 2]$ has :

- (1) 8 as absolute maximum value
 (2) 4 as absolute maximum value
 (3) 4 as absolute minimum value
 (4) -3 as absolute minimum value



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (1, 3)
3. (3)
4. (3)
5. (1)
6. (3)
7. (4)

8. (1)
9. (2)
10. (1)
11. (1)
12. (1)
13. (1, 4)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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